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Master Thesis

Design and development of Seebeck coefficient and electrical conductivity integrated instrument for flexible materials

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CHAPTER -1

Introduction

1.1 Background

In recent times, the field of Flexible and wearable thermoelectric devices has shown great promise in research and development, owing to its potential applications in energy harvesting, personalized temperature control, and wearable electronics. These devices are designed to convert temperature differences into electrical energy efficiently, providing a sustainable and convenient power source for various wearable applications. One of the significant advantages of flexible thermoelectric devices lies in their ability to adapt to irregular shapes and seamlessly integrate with wearable garments and accessories. This characteristic ensures comfortable and unobtrusive usage as they can be incorporated into clothing like jackets, shirts, or socks, or even attached to body parts to harness the temperature disparity between the body and the surrounding environment. The materials utilized in these devices include polymers, composites, conductive fabrics, and thin films, which possess unique mechanical and electrical properties suitable for wearable applications. These materials offer excellent flexibility and mechanical durability and enable the devices to withstand the daily stresses and movements encountered by users. Additionally, advancements in materials science and nanotechnology have further enhanced the performance and efficiency of flexible thermoelectric materials.

The design process of flexible and wearable thermoelectric devices involves the integration of multiple components, such as thermoelectric materials, electrodes, substrates, and electrical interconnects, to create a flexible and robust structure. This requires careful consideration of factors like material compatibility, mechanical flexibility, and thermal management. Various fabrication techniques like printing, coating, and thin-film deposition are employed to achieve the desired device architecture while maintaining flexibility.

Apart from energy harvesting, wearable thermoelectric devices offer the potential for personalized climate control. By utilizing temperature differences between the body and the environment, these devices can provide localized heating or cooling to enhance comfort and regulate thermal conditions. This opens up new avenues for applications in smart textiles, sports apparel, and medical garments, where personalized and localized thermal management is highly desirable. However, despite the promising progress in the field of flexible and wearable thermoelectric devices, several challenges persist. These challenges include enhancing the efficiency and power output of the devices, optimizing the flexibility and durability of materials, improving compatibility with textile substrates, and developing effective heat management strategies. Overcoming these hurdles will require collaborative efforts involving materials scientists, engineers, and designers to advance the field and fully exploit the potential of flexible and wearable thermoelectric devices. This technology holds great promise for further development, given the continuous heat emission from the human body, which maintains a stable temperature of approximately 36.5 °C [Fig-1]. As the human body operates with a mechanical efficiency of only around 15-30%, a significant portion of energy, around 70-75%, is dissipated in the form of heat. It is, therefore, reasonable to explore the feasibility of capturing this abundant thermal energy and converting it into valuable electrical energy. By Making a closely observing the daily activities of individuals, it becomes evident that a substantial amount of energy is generated through routine tasks and physical activities [Fig-2].

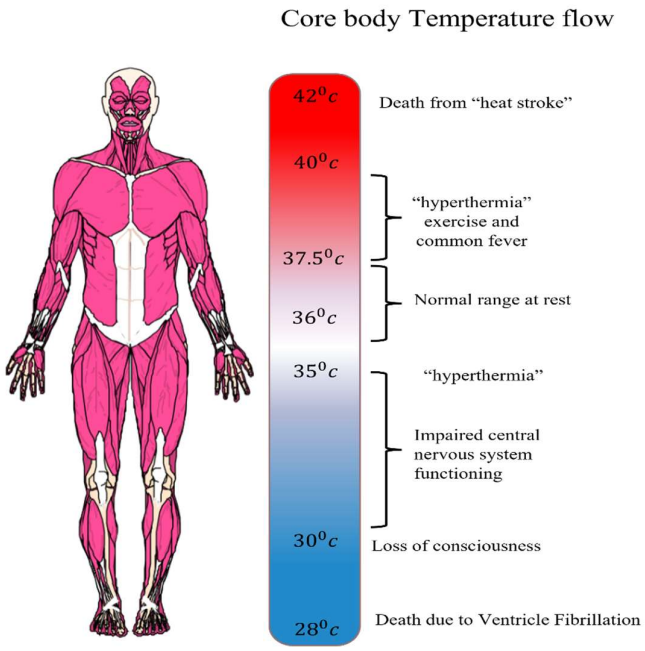


Figure -1 Core body temperature flow in various parts o the human body

Daily Activity	WATTS	Kilo Cal/hr
Sleeping	81	70
Sitting	93	80
Standing	116	110
Conversation	128	110
Eating	128	110
Driving	163	140
House keeping	175	150
Swimming	582	500
Running	1048	900

Table 1 Power enhancement by the moment of body.

1.2 Objectives

Flexible thermoelectric materials and devices have emerged as highly promising candidates for various applications due to their unique mechanical properties, lightweight nature, and flexibility. These materials and devices find practical use in diverse fields such as wearable electronics, energy harvesting, and flexible sensors. To optimize their performance in these applications, it is essential to accurately characterize their thermoelectric and electrical properties. Two fundamental parameters crucial in this regard are the Seebeck coefficient and electrical conductivity. The Seebeck coefficient quantifies a material's ability to generate a voltage in response to a temperature gradient, making it a key factor in energy conversion applications like thermoelectric generators. Conversely, electrical conductivity assesses a material's capacity to conduct electric current, playing a critical role in evaluating its suitability for electronic devices.

Existing techniques for measuring the Seebeck coefficient and electrical conductivity often require separate instruments, lacking integration and flexibility. Additionally, these techniques may not be ideally tailored for flexible materials, which pose unique challenges such as conformability and varying sample sizes. To overcome these limitations, our research focuses on designing and developing an integrated instrument capable of measuring both the Seebeck coefficient and electrical conductivity of flexible materials. This integrated instrument aims to provide researchers and engineers with a comprehensive tool for efficient and reliable measurement and analysis. By incorporating temperature control, signal conditioning, and data acquisition functionalities, the instrument ensures accurate measurements across diverse environmental conditions.

The design of the integrated instrument considers various factors, including the measurement range and resolution to capture fine-grained variations in the Seebeck

coefficient and electrical conductivity. It also features robust temperature control mechanisms to ensure stability during measurements and flexible adaptability to different sample sizes and shapes. Moreover, user-friendliness is a priority, with an intuitive interface for effortless operation, data acquisition, and analysis, coupled with safety considerations to safeguard both users and the instrument itself. The development of this integrated instrument carries significant implications for the flexible materials research field, empowering researchers with a powerful tool to optimize the thermoelectric and electrical performance of these materials. Furthermore, the instrument's capabilities have the potential to stimulate innovation by enabling the exploration of new materials and designs to enhance thermoelectric and electrical properties in the realm of flexible materials.

Our research objectives encompass the design and development of the integrated instrument, its validation through experimental measurements, and the exploration of potential applications in the field. This work covers hardware design, software development, experimental setup, and the meticulous analysis of the instrument's performance.

1.3 Scope

This project is dedicated to developing a customized integrated instrument specifically designed for characterizing flexible materials. The instrument's primary functionalities involve measuring the Seebeck coefficient and electrical conductivity within a specified temperature range. The scope of the project covers hardware design, software development, experimental validation, and comprehensive performance analysis of the instrument. However, it is important to note that the fabrication or synthesis of flexible materials is not part of the project's scope, as the instrument's purpose is to analyze pre-existing samples. The instrument's capabilities extend to various types of flexible materials, including polymers, composites, and thin films. The overarching goal of this project is to advance the field of flexible materials

research and offer researchers and engineers a valuable tool for optimizing the thermoelectric and electrical properties of these materials.

CHAPTER -2

SEEBECK COEFFICIENT AND ELECTRICAL CONDUCTIVITY

2.1 Seebeck Coefficient

The Seebeck coefficient is a fundamental parameter that characterizes a material's thermoelectric properties. It quantifies the voltage generated across the material when subjected to a temperature gradient. This phenomenon, known as the Seebeck effect, was discovered by physicist Thomas Johann Seebeck in 1821. The basic principle behind the Seebeck effect involves the interplay between temperature gradients and charge carriers within the material. In a material with varying temperatures at its ends, the charge carriers (such as electrons or holes) experience different levels of thermal energy. As a result, these charge carriers tend to migrate from the hotter region to the cooler region, creating a voltage difference along the material. This voltage difference, termed the Seebeck voltage, is directly proportional to the temperature gradient and can be measured across the material using electrodes.

The Seebeck coefficient is mathematically defined as the ratio of the generated voltage to the temperature difference across the material. It serves as a metric for the material's capacity to convert thermal energy into electrical energy. Typically represented by the symbol "S," the Seebeck coefficient is measured in units of volts per degree Kelvin (V/K) or volts per degree Celsius (V/°C). The sign of the Seebeck coefficient indicates the polarity of the voltage generated concerning the temperature gradient.

Several factors influence the Seebeck coefficient, including the material type, crystal structure, and characteristics of the charge carriers. Different materials exhibit distinct Seebeck coefficients due to variations in their electronic band structures and carrier transport properties. Thermoelectric applications favor materials with high absolute values of the Seebeck coefficient, as they can generate larger voltages for a given temperature gradient.

Accurate measurement of the Seebeck coefficient is crucial for effectively characterizing and optimizing thermoelectric materials. It provides valuable insights into a material's suitability for thermoelectric applications and its potential to convert

waste heat into usable electrical energy. Various experimental techniques, such as the differential thermocouple method or the steady-state method, are employed to measure the Seebeck coefficient with precision. These techniques involve precise temperature control, calibration, and the use of reference materials to ensure reliable and reproducible results.

Measuring the Seebeck coefficient in flexible thermoelectric devices necessitates specialized techniques that account for the unique properties and structures of flexible materials. Here are several commonly employed methods for measuring the Seebeck coefficient in such materials:

2.1.1. Differential Temperature Method: The differential temperature method is a widely used technique for measuring the Seebeck coefficient. It involves placing a flexible thermoelectric sample between two temperature-controlled platforms. The temperature gradient is applied across the sample, and the resulting voltage difference is measured using electrodes. By varying the temperature difference and measuring the corresponding voltage, the Seebeck coefficient can be determined.

2.1.2 Harman Method: The Harman method, also known as the two-heater method, is a variation of the differential temperature method. It involves using two separate heaters to create a temperature gradient across the flexible thermoelectric sample. The voltage generated across the sample is measured using electrodes, and the Seebeck coefficient is calculated based on the temperature difference and the voltage measurement.

2.1.3. Transient Thermoelectric Technique: The transient thermoelectric technique is a dynamic method for measuring the Seebeck coefficient. It involves applying a step change in temperature to the flexible thermoelectric sample and monitoring the subsequent voltage response. By analyzing the transient voltage data, the Seebeck coefficient can be determined. This method is particularly useful for characterizing the time-dependent behavior of the Seebeck coefficient.

2.1.4 Kelvin Probe Method: The Kelvin probe method, which is commonly used to measure the work function difference between materials, can also be adapted to measure the Seebeck coefficient. By creating a temperature gradient across the flexible thermoelectric sample and using a Kelvin probe to measure the contact potential difference, the Seebeck coefficient can be calculated based on the temperature difference and the voltage measurement.

2.1.5 Harman 4-Probe Method: The Harman 4-probe method is a more advanced technique for measuring the Seebeck coefficient in flexible thermoelectric devices. It involves using four electrodes to measure the voltage drop along the length of the sample. By applying a known temperature gradient and measuring the voltage at multiple positions, the Seebeck coefficient can be determined more accurately, taking into account any variations in the material's properties.

2.1.6 Impedance Spectroscopy: While primarily used for electrical conductivity measurements, impedance spectroscopy can also provide information about the Seebeck coefficient. By analyzing the frequency-dependent impedance response of the flexible thermoelectric sample, the complex Seebeck coefficient, which includes both the magnitude and phase information, can be extracted.

These measurement techniques offer various approaches to determining the Seebeck coefficient in flexible thermoelectric devices. The choice of method depends on factors such as the specific requirements of the experiment, the nature of the sample, and the desired accuracy. Researchers often employ a combination of techniques to obtain a comprehensive understanding of the Seebeck coefficient in flexible thermoelectric materials.

2.2 Electrical Conductivity

The electrical conductivity plays a vital role in thermoelectric devices as it determines a material's ability to carry electrical current. High electrical conductivity is desirable

in these devices to effectively convert thermal energy into electrical energy. This section explores the fundamental concepts of electrical conductivity in thermoelectric devices. It represents how easily electrical current can flow through a substance and measures the material's capability to transport charge carriers (such as electrons or holes) in the presence of an electric field. For thermoelectric devices, electrical conductivity is crucial for facilitating electrical current flow across the device, leading to electricity generation. The electrical conductivity of a material is influenced by various factors, including the concentration of charge carriers, which is controlled by the material's doping level or intrinsic properties. Doping introduces additional charge carriers into the material, enhancing its electrical conductivity. Another significant factor affecting electrical conductivity is the mobility of the charge carriers, measuring their ability to move in response to an electric field. Materials with high carrier mobility exhibit higher electrical conductivity.

In thermoelectric devices, a desirable scenario involves achieving high electrical conductivity while maintaining low thermal conductivity. This combination ensures efficient electrical conduction while minimizing heat loss, ultimately maximizing the thermoelectric conversion efficiency. However, this balance can be challenging to achieve, as increasing the doping level to enhance conductivity may inadvertently raise the material's thermal conductivity. Therefore, researchers adopt various techniques to accurately measure the electrical conductivity of thermoelectric materials. These methods involve applying an electric field to the material and measuring the resulting current, which is then used to calculate the electrical conductivity based on the material's geometry and dimensions. Additionally, measurements are conducted at different temperatures to assess the temperature dependence of electrical conductivity, crucial for understanding the material's behavior in thermoelectric applications. To enhance electrical conductivity in thermoelectric materials, researchers focus on optimizing several factors. These include selecting materials with high carrier mobility, controlling the doping level to

achieve the desired carrier concentration, and engineering the material's microstructure to facilitate efficient charge carrier transport. Furthermore, strategies such as Nano structuring, alloying, and incorporating dopants can further improve electrical conductivity in thermoelectric materials.

Regarding flexible thermoelectric devices, there are various methods for measuring electrical conductivity. These techniques differ in principle, complexity, and the information they provide. Here are some commonly used methods for measuring electrical conductivity in flexible thermoelectric devices:

2.2.1 Four-Point Probe Method: The four-point probe method is a widely used technique for measuring electrical conductivity. It involves placing four evenly spaced electrical probes on the surface of the material in a linear arrangement. A known current is passed through the outer probes, while the voltage drop across the inner probes is measured. By applying Ohm's law, the electrical conductivity of the material can be determined.

2.2.2 Van der Pauw Method: The Van der Pauw method is another popular technique for measuring electrical conductivity. It involves creating a circular or square sample with contacts placed at four equidistant points along the perimeter. The current is passed through one pair of contacts, and the voltage is measured across the other pair. By varying the current and voltage measurements between different contact pairs, the electrical conductivity can be calculated using the Van der Pauw equation.

2.2.3 Hall Effect Measurement: The Hall effect is utilized to measure electrical conductivity in the presence of a magnetic field. A thin strip of flexible thermoelectric material is used, and a magnetic field is applied perpendicular to the current flow. By measuring the Hall voltage, which is generated perpendicular to both the current and the magnetic field, the electrical conductivity and carrier concentration of the material can be determined.

2.2.4 Impedance Spectroscopy: Impedance spectroscopy is a technique that measures the electrical response of a material over a range of frequencies. It involves applying an AC voltage signal to the material and measuring the resulting current. By analyzing the impedance data, which includes both the resistive and capacitive components, the electrical conductivity can be extracted.

2.2.5 Transmission Line Method: The transmission line method involves fabricating a transmission line structure on the flexible thermoelectric material. The electrical conductivity is determined by measuring the impedance or the transmission coefficient of the transmission line. This method is particularly useful for thin film or nanowire-based thermoelectric materials.

2.2.6 Kelvin Probe Method: The Kelvin probe method, also known as the contact potential difference method, measures the work function difference between two materials in contact. By placing a reference material with a known work function in contact with the flexible thermoelectric material, the contact potential difference can be measured. This measurement can be correlated to the electrical conductivity of the thermoelectric material.

Each of these methods has its advantages and limitations, and the choice of measurement technique depends on factors such as the material type, sample size, accuracy requirements, and experimental setup. Researchers often employ a combination of these techniques to gain a comprehensive understanding of the electrical conductivity in flexible thermoelectric devices.

2.3 Flexible Materials

Flexible materials, including conductive fabrics, have gained significant interest in recent years due to their unique properties and wide range of applications. These materials offer mechanical flexibility, lightweight characteristics, and the ability to conform to different shapes and surfaces. Conductive fabrics, in particular, possess electrical conductivity while retaining flexibility and textile-like properties, making

them suitable for applications in wearable electronics, smart textiles, and electronic textiles. The basic principle behind flexible conductive fabrics lies in their structure and composition. These fabrics are typically composed of conductive fibers or yarns interwoven with non-conductive fibers or materials. The conductive fibers can be made from various materials, such as metal-coated fibers, carbon nanotubes, or conductive polymers. The non-conductive fibers provide flexibility and mechanical strength to the fabric. By combining these elements, conductive fabrics exhibit electrical conductivity while retaining the desired flexibility and textile-like characteristics. The electrical conductivity of conductive fabrics arises from the presence of conductive pathways within the fabric structure. The conductive fibers or coatings enable the flow of electric current through the fabric, allowing for the transmission of signals or power. The conductivity of the fabric can be controlled by factors such as the type and concentration of conductive materials used, the weaving pattern, and the fabric's thickness. Flexible conductive fabrics find applications in various fields. In wearable electronics, these fabrics can be used to create integrated circuits, sensors, and electrodes within clothing or accessories. They enable the development of comfortable and unobtrusive wearable devices that can monitor vital signs, track movement, or facilitate communication. In smart textiles, conductive fabrics can be employed to create interactive garments or surfaces that respond to touch, pressure, or temperature. They can also be utilized in electronic textiles for applications such as electromagnetic shielding or heating elements.

The development and advancement of flexible conductive fabrics require interdisciplinary efforts combining materials science, textile engineering, and electronics. Researchers and engineers work on improving the conductivity, durability, and washability of these fabrics while ensuring their comfort and flexibility. Novel fabrication techniques, such as additive manufacturing or printed electronics, are being explored to create complex patterns and circuits directly onto the fabric, further expanding the capabilities of flexible conductive fabrics.

After conducting an in-depth study on the Seebeck coefficient and electrical conductivity of different flexible materials, we have decided to focus our efforts on designing and developing an integrated instrument. This instrument will utilize a combination of the differential temperature method and the four-point probe method[Fig-23] to accurately measure and analyze key parameters such as the Seebeck coefficient and electrical conductivity specifically for conductive fabrics. To obtain precise and reliable measurements, the differential temperature method will enable us to measure the voltage generated across the conductive fabric sample due to a controlled temperature gradient. By carefully controlling the temperature difference and monitoring the corresponding voltage, we can determine the Seebeck coefficient, which quantifies the material's ability to convert thermal energy into electrical energy. In addition to the Seebeck coefficient, we will also employ the four-point probe method to measure the electrical conductivity of the conductive fabric. This technique involves placing four electrically isolated probes onto the fabric surface, applying a known current through the outer probes, and measuring the voltage drop across the inner probes. By analyzing the resistance and geometric properties of the sample, we can accurately determine the electrical conductivity, which reflects the material's ability to conduct electric current. The integration of these two methods into a single instrument will allow us to streamline the measurement process and obtain comprehensive data on the thermoelectric and electrical properties of conductive fabrics. The instrument will incorporate precise temperature control mechanisms, advanced signal conditioning techniques, and high-precision data acquisition to ensure accurate and repeatable measurements. The development of a user-friendly interface and software will facilitate data analysis and provide researchers and engineers with valuable insights for further optimization of conductive fabrics in wearable electronics and related applications.

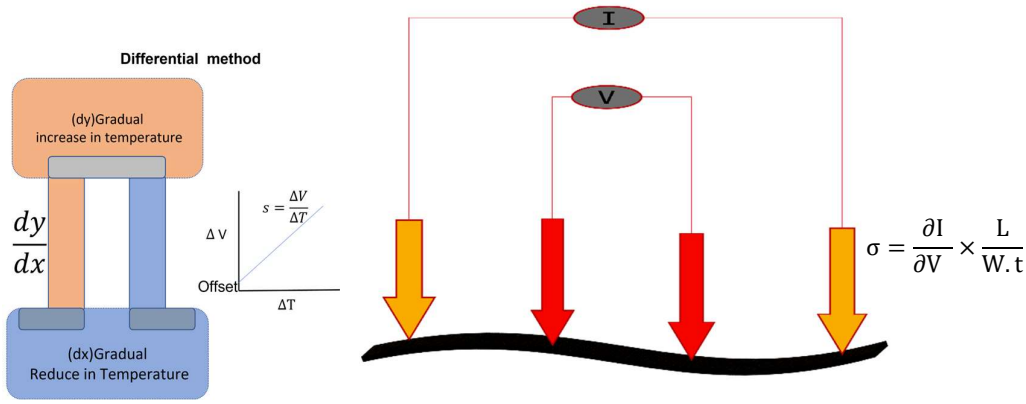


Figure 2 Schematic view of the differential method and four probe method.

By employing the differential temperature method in combination with the four-point probe method, our integrated instrument will enable researchers to explore the relationship between the Seebeck coefficient and the electrical conductivity of conductive fabrics. This research will contribute to the advancement of flexible material characterization and prove the way for the development of tailored instruments specifically designed for the analysis and optimization of thermoelectric and electrical properties in conductive fabrics.

CHAPTER -3

Design Considerations

3.1 Measurement Considerations:

The main objective of our project was to develop an integrated instrument capable of measuring both the Seebeck coefficient and electrical conductivity of flexible materials, with a specific focus on conductive fabrics. We are in the process of measuring a 5x2cm^2 area of the flexible material. We have conducted several design iterations [Fig- 3,4,5,6,7,8] to establish the optimal interaction between the fabric and the probes.

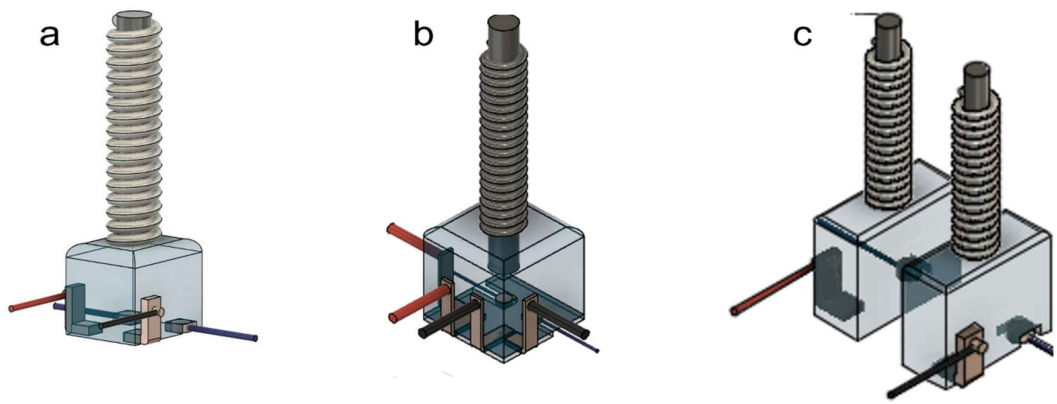


Figure -3. a,b,c Different possible designs for the integration probe

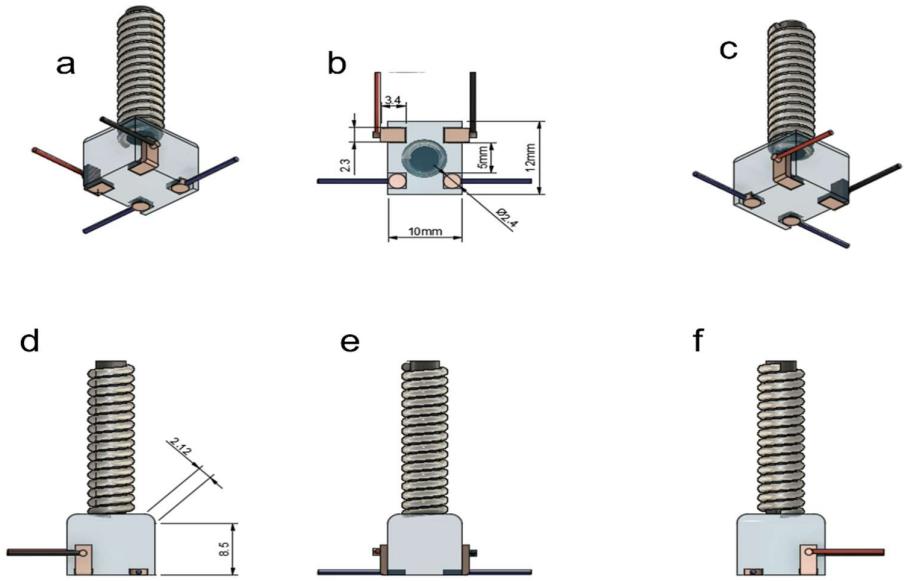


Figure 4.5 (a) Bottom -right (b) bottom (c) Bottom -left (d)left side (e) back

(f) the right side of the design a-type integrated design probe in Fig. 3.

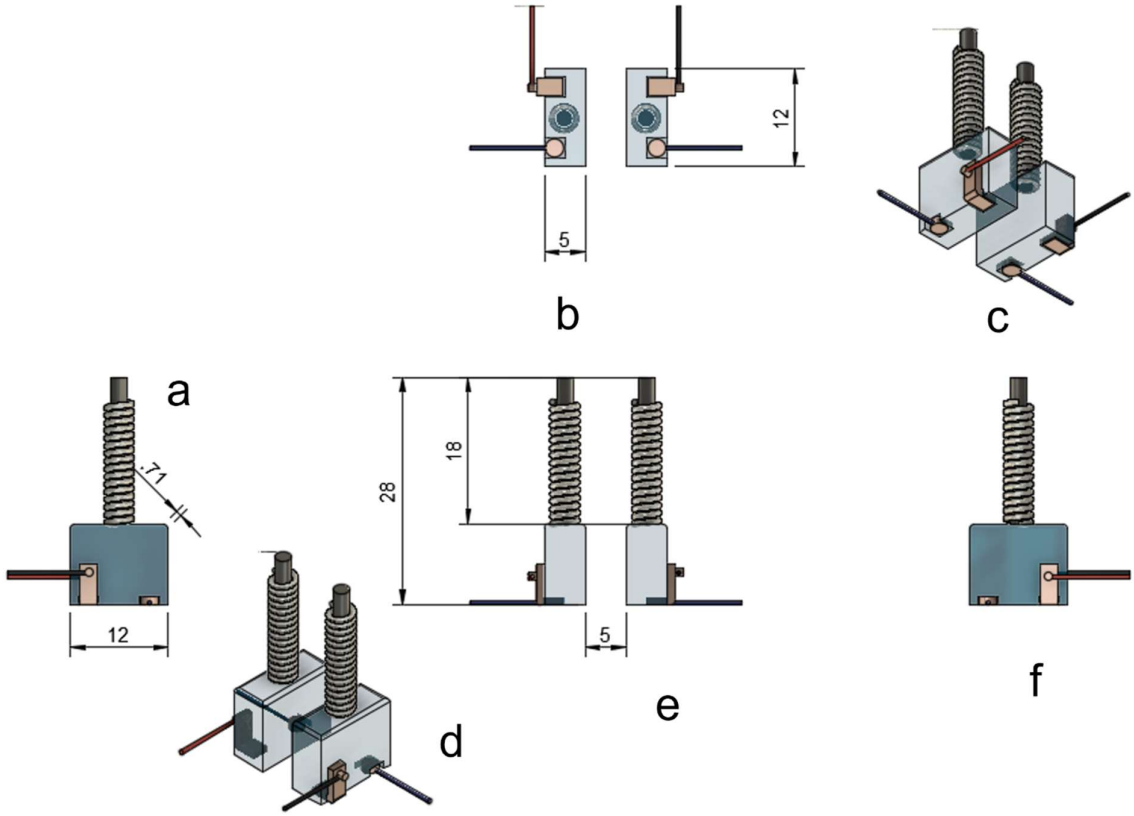


Figure -5. (a) left side (b) bottom (c) Bottom -left (d)Top right (e) back (f) right side of the design c-type integrated design probe in Fig. 3.

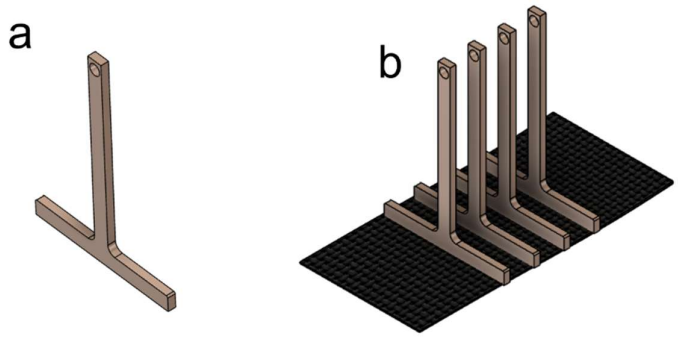


Figure 6 .7 (a) copper probe design for the fabric samples (b) integrated probes arrangement on the carbon fabric in Fig. 3.

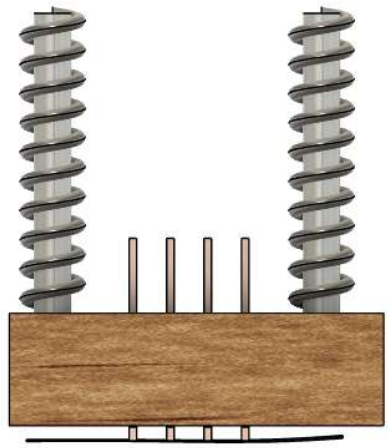


Figure -7. single integrated probe with the holder set on the carbon fabric(CAF).

Our aim was to provide researchers and engineers with a comprehensive tool for accurately characterizing and optimizing these materials for various applications. To achieve this goal, we prioritized the needs of our users. The instrument features a user-friendly interface, making it easy to operate and control. Researchers can effortlessly switch between different measurement modes and access real-time data visualization for effective analysis. The software interface allows for customizable settings, ensuring adaptability to different research requirements.

3.2 Instrumentation Requirements

Regarding the hardware design, our primary focus was on achieving precision and reliability. The instrument incorporates advanced temperature control mechanisms to maintain a stable and controlled thermal environment during measurements. This is crucial for obtaining accurate Seebeck coefficient values. Additionally, our four-point probe system utilizes high-quality materials and precise alignment to ensure precise electrical conductivity measurements on conductive fabrics. The probes are designed to minimize interference and establish reliable contact with the fabric surface. The

instrument's compact and portable design provides flexibility in usage, allowing researchers to conduct measurements both in the laboratory and in the field. We recognize the importance of adaptability when working with flexible materials, and thus, our instrument can accommodate various sample sizes and shapes. Whether dealing with a small fabric patch or a large wearable device, our instrument can reliably measure the thermoelectric and electrical properties of a wide range of flexible materials. Safety was also a paramount consideration in our design process. The instrument integrates built-in safety features to safeguard both the user and the instrument itself from potential hazards during operation. Thorough testing has been conducted, and we have adhered to industry standards to ensure the highest level of safety and reliability.

Moreover, we underwent multiple iterations of the designs to maximize the coverage area of the fabric since conductive fabric often exhibits unevenness due to its yarning pattern. Ultimately, we successfully developed a suitable probe that efficiently covers the fabric's surface.

3.3 Integration for Flexible Materials (conductive fabrics)

The integration of these methods with conductive fabrics opens up fascinating possibilities for leveraging the thermoelectric and electrical properties of these materials. Through the creation of an integrated instrument that combines the differential temperature method and the four-point probe method, we gain the ability to accurately measure and analyze the Seebeck coefficient and electrical conductivity of conductive fabrics.

Firstly, the differential temperature method can be seamlessly integrated into the design of flexible conductive fabrics. By incorporating temperature-controlling elements within the fabric structure, we can create controlled temperature gradients across the material. These gradients will induce the Seebeck effect, generating voltage differences that can be precisely measured using integrated electrodes. This innovative

approach enables real-time monitoring of the thermoelectric properties of the fabric as it interacts with the surrounding environment and the human body, offering valuable insights into its energy conversion capabilities.

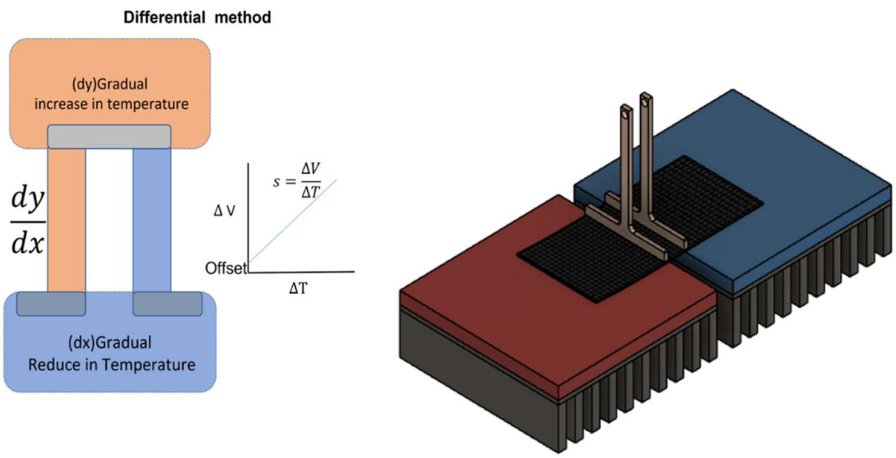
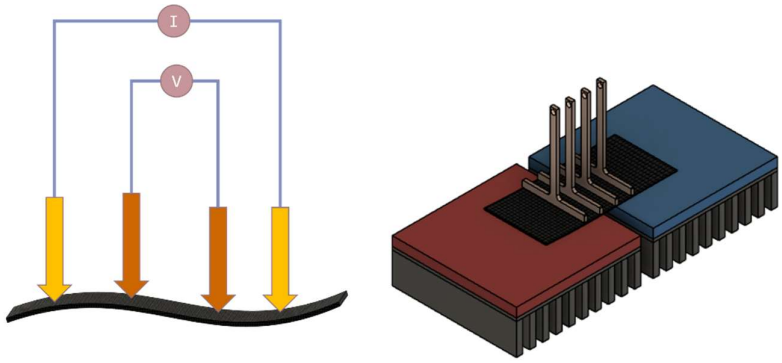


Figure -8 Differential temperature method of the fabrics with incorporating temperature-controlling elements.

Secondly, the four-point probe method can be adapted for use with flexible conductive fabrics. By designing flexible probes that can conform to the fabric's surface, we can establish stable electrical contact points. The integration of these probes into the fabric structure ensures minimal interference during electrical conductivity measurements. This allows us to accurately assess the fabric's electrical conductivity, reflecting its ability to conduct electric current and provide valuable information for optimizing the material's electrical performance.



$$\sigma = \frac{\partial I}{\partial V} \times \frac{L}{W.t}$$

Figure -9 Four-point probe method of the fabrics incorporating temperature-controlling elements.

To fully leverage the potential of flexible conductive fabrics, the integrated instrument's hardware design should consider the unique mechanical properties and conformability of these materials. The use of flexible substrates, conductive yarns, or nanocomposite materials can enhance the instrument's adaptability to the fabric's characteristics, ensuring reliable measurements across various fabric shapes and sizes. The instrument's software interface should be tailored for easy operation and data analysis with clear visualization of the Seebeck coefficient and electrical conductivity results. Researchers and engineers can benefit from customizable settings to accommodate different fabric types and optimize energy conversion efficiency. To fully leverage the potential of flexible conductive fabrics, the integrated instrument's hardware design should consider the unique mechanical properties and conformability of these materials. The use of flexible substrates, conductive yarns, or nanocomposite materials can enhance the instrument's adaptability to the fabric's characteristics, ensuring reliable measurements across various fabric shapes and sizes.

The instrument's software interface should be tailored for easy operation and data analysis with clear visualization of the Seebeck coefficient and electrical conductivity results. Researchers and engineers can benefit from customizable settings to accommodate different fabric types and optimize energy conversion efficiency.

CHAPTER -4

Hardware Design& Software Development

4.1 Instrument Block Diagram

The instrumental block diagram for designing and developing an integrated instrument for the Seebeck coefficient and electrical conductivity of flexible materials showcases an advanced and multifunctional system. It combines two crucial measurement techniques. The differential temperature method [fig-10,11]and the four-point probe method[fig-12,13]. The core element of the instrument is the temperature control module, which utilizes precise sensors and heating elements to establish controlled temperature gradients within the flexible material samples. These controlled gradients induce the Seebeck effect, leading to the generation of voltage differences within the material.

Differential temperature method

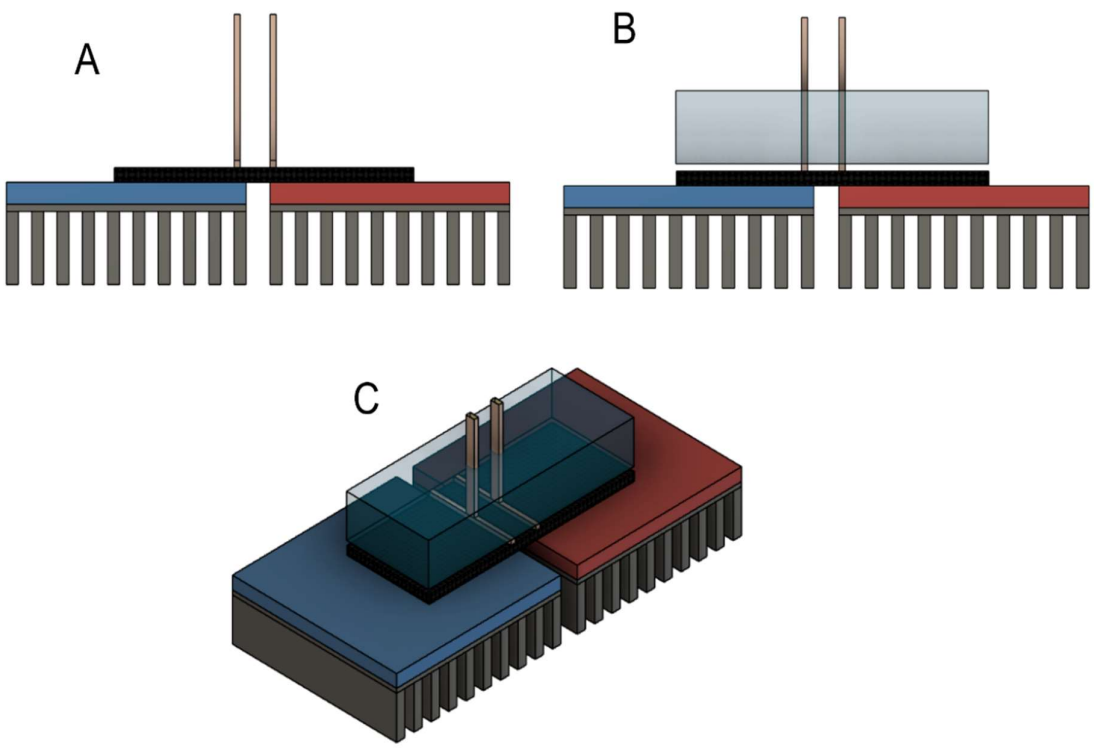


Figure-10 A,B,C Probe arrangement on the Sample for the differential temperature method.

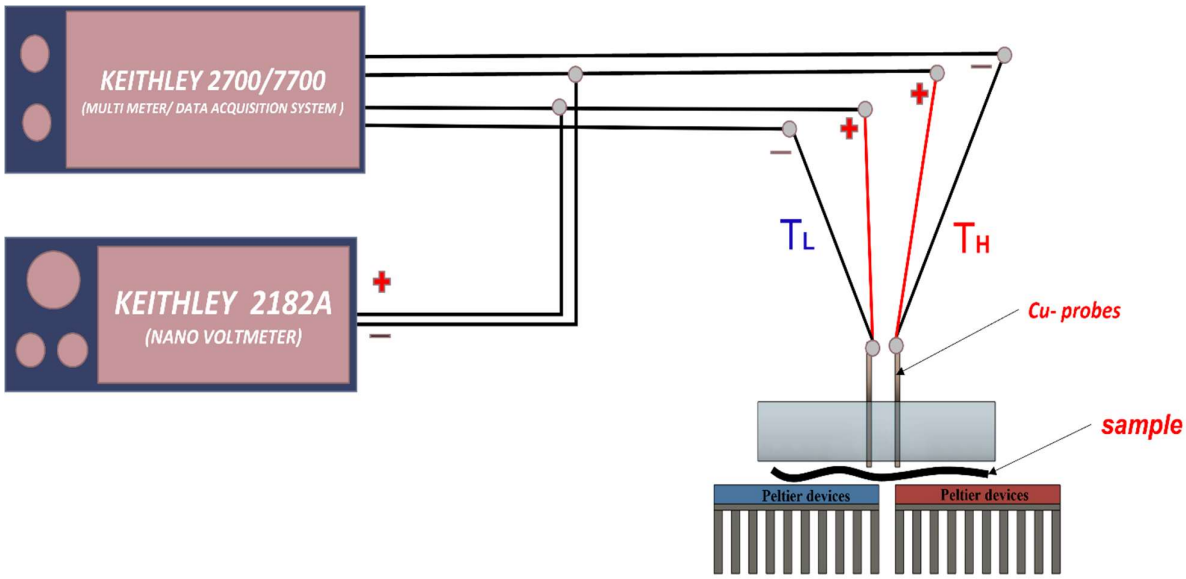


Figure -11 Schematic block diagram of the differential temperature method.

Four-point probe system

Another essential component is the four-point probe system, [Fig-12] specially designed for accurate electrical conductivity measurements on conductive fabrics. High-quality probes ensure reliable contact with the fabric surface, minimizing interference and ensuring consistent and reproducible data. The electrodes within the fabric connect to the data acquisition unit[Fig-13], facilitating real-time measurement of the voltage.

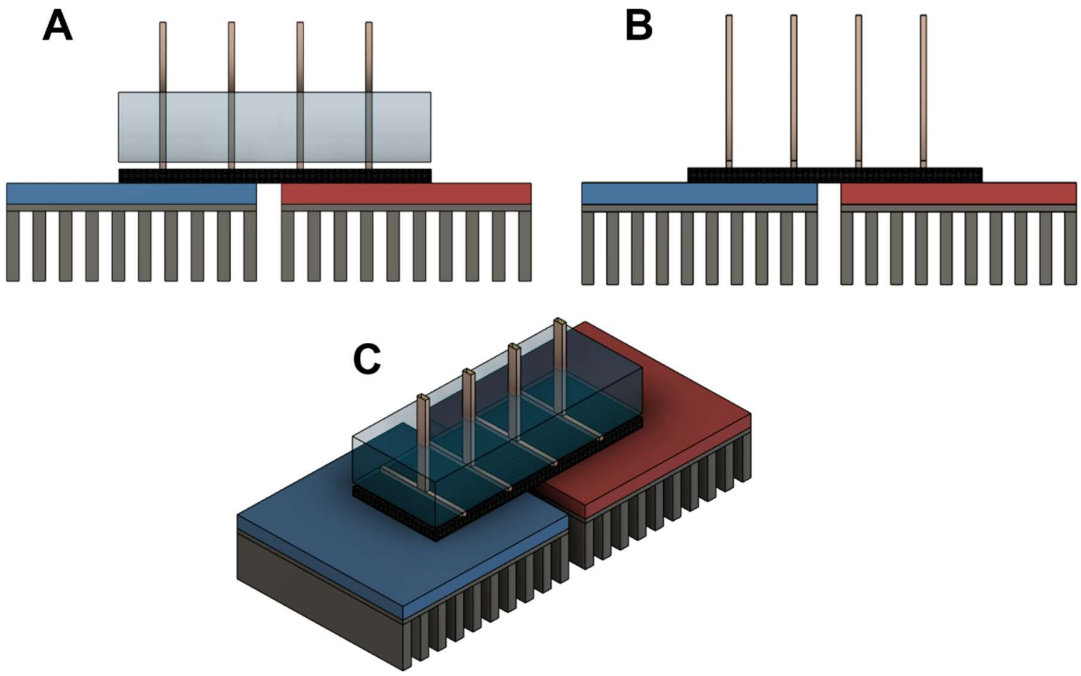


Figure-12. A,B,C Probe arrangement on the Sample for the four-point probe method.

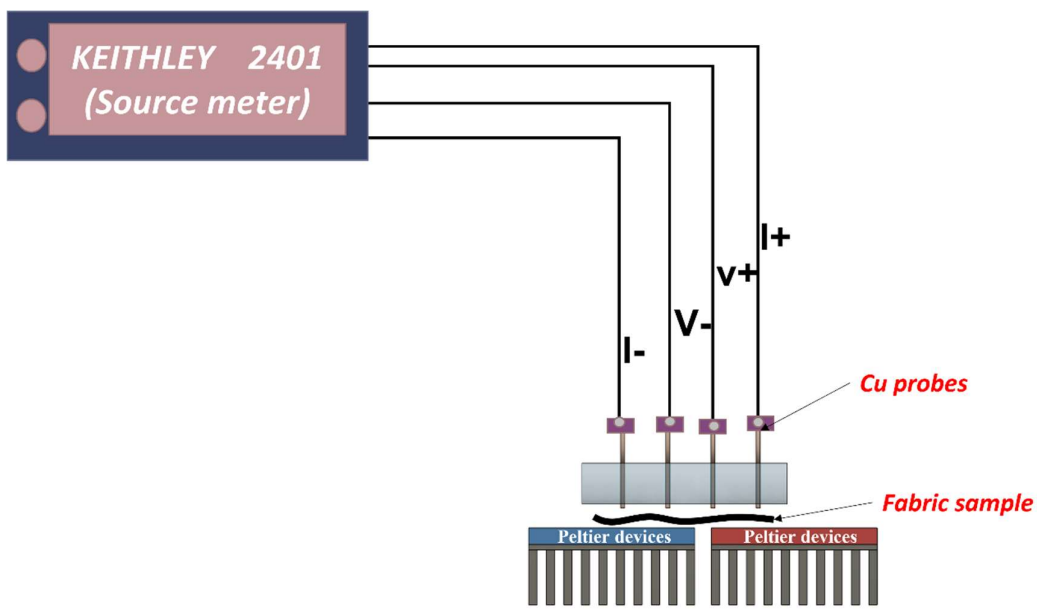


Figure- 13 Schematic block diagram of the four-point probe system.

4.2 Single Integrated Probe

The Single Integrated Probe is a novel add-on feature of our instrument, designed to enhance the measurement capabilities for flexible materials. This integrated probe combines the functionalities of both the differential temperature method and the four-point probe method into a single, compact unit. With this new probe[Fig-14,15], researchers can efficiently measure both the Seebeck coefficient and electrical conductivity of flexible materials without the need for separate instruments or multiple setups. The Single Integrated Probe is equipped with precise temperature-controlling elements that can create controlled temperature gradients across the flexible material. These gradients induce the Seebeck effect, generating voltage differences that can be accurately measured using the Keithley 2182A system. This seamless integration of temperature control by Keithley 2182A, 2700 and electrical measurements Keithley 2401 by ensures real-time monitoring[Fig-14,16,17] of the thermoelectric properties of the material, providing valuable insights into its energy conversion capabilities.

Furthermore, the Single Integrated Probe is designed with adaptability in mind, allowing for measurements on various sample sizes within (5x2)cm^2 and shapes of conductive fabrics. It's high-quality materials and precise alignment ensure reliable contact with the fabric surface, minimizing interference and optimizing measurement accuracy. With the Single Integrated Probe, researchers and engineers can efficiently characterize and optimize the thermoelectric and electrical properties of flexible materials, unlocking their full potential for a wide range of applications in wearable electronics, energy harvesting, and flexible sensors.

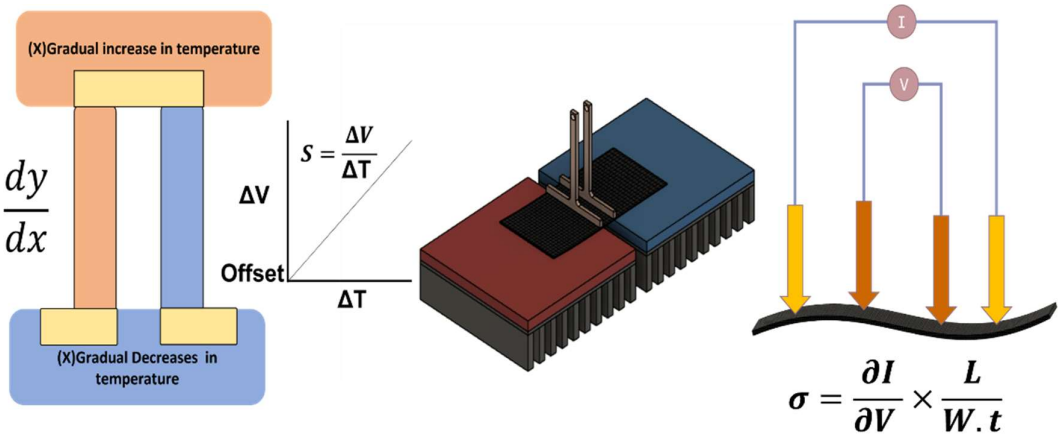


Figure-14. Single Probe arrangement on the Sample for the both differential temperature method and four-point probe method

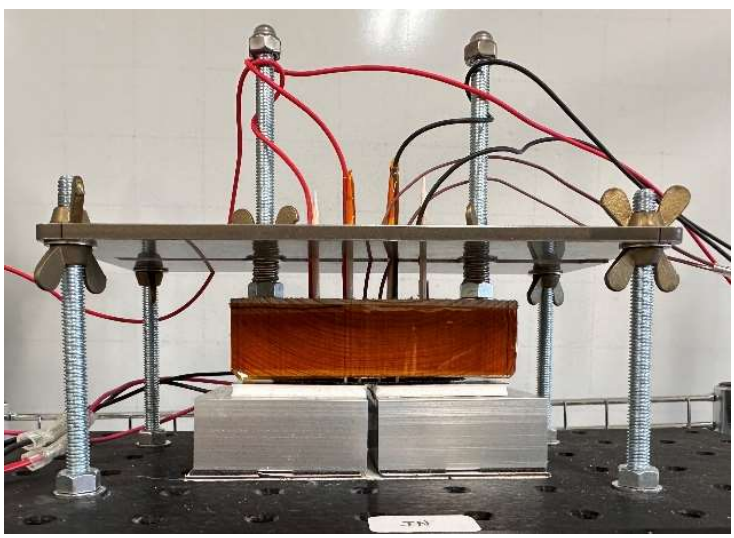


Figure-15. Single integrated Probe prototype on the Sample for the both differential temperature method and four-point probe method.

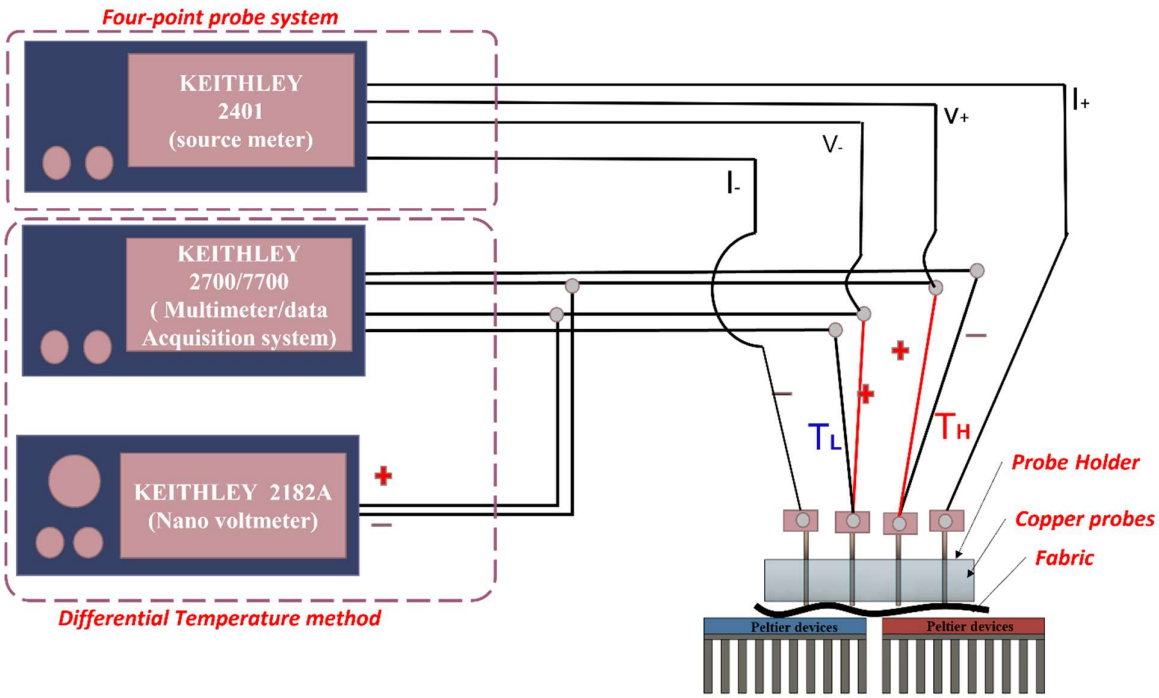


Figure- 16 Schematic block diagram of the single integrated probe.

4.3 Data Acquisition System

The Data Acquisition System implemented in this study is designed to facilitate accurate temperature control using the Matsuda power supply while simultaneously collecting essential data through the Keithley 2182A, 2700, and Keithley 2401 instruments.

The Matsuda power supply serves as the primary temperature-controlling system, enabling the creation of controlled temperature gradients across to Peltier module for flexible materials. By precisely regulating the temperature, the power supply allows researchers to induce the Seebeck effect within the materials, leading to the generation of voltage differences that can be measured using the integrated Keithley instruments. The Keithley 2182A nanovoltmeter is a critical component of the Data Acquisition System, responsible for maintaining a stable thermal environment during measurements. This nanovoltmeter's exceptional sensitivity ensures the capture of extremely low voltage signals, which is crucial for accurately measuring the subtle

voltage variations induced by the Seebeck effect in flexible materials. By using the Keithley 2182A alongside the Matsuda power supply, researchers can obtain precise Seebeck coefficient values, providing valuable insights into the thermoelectric properties of the materials

Alongside temperature control, the Data Acquisition System incorporates the Keithley 2700 multimeter, an instrument capable of seamless integration with the fabric structure. The Keithley 2700 allows for real-time monitoring of the thermoelectric properties of the flexible materials, complementing the Seebeck coefficient measurements. Its versatile measurement modes and user-friendly interface offer researchers the flexibility to adapt the system to their specific research requirements, while the real-time data visualization capabilities facilitate efficient data analysis.

The Data Acquisition System incorporates the Keithley 2401 source-meter unit for assessing the electrical conductivity of conductive fabrics. The Keithley 2401's four-point probe system ensures accurate electrical measurements by minimizing interference and providing reliable contact with the fabric surface. By integrating the Keithley 2401 alongside the temperature-controlled setup, researchers can comprehensively evaluate both the thermoelectric and electrical properties of the flexible materials, enabling a thorough understanding of their energy conversion capabilities. The synergy of the Matsuda power supply, Keithley 2182A, 2700, and 2401 instruments within the Data Acquisition System creates a robust platform for characterizing flexible materials. This integrated system empowers researchers and engineers to efficiently analyze and optimize the thermoelectric and electrical properties of flexible materials.

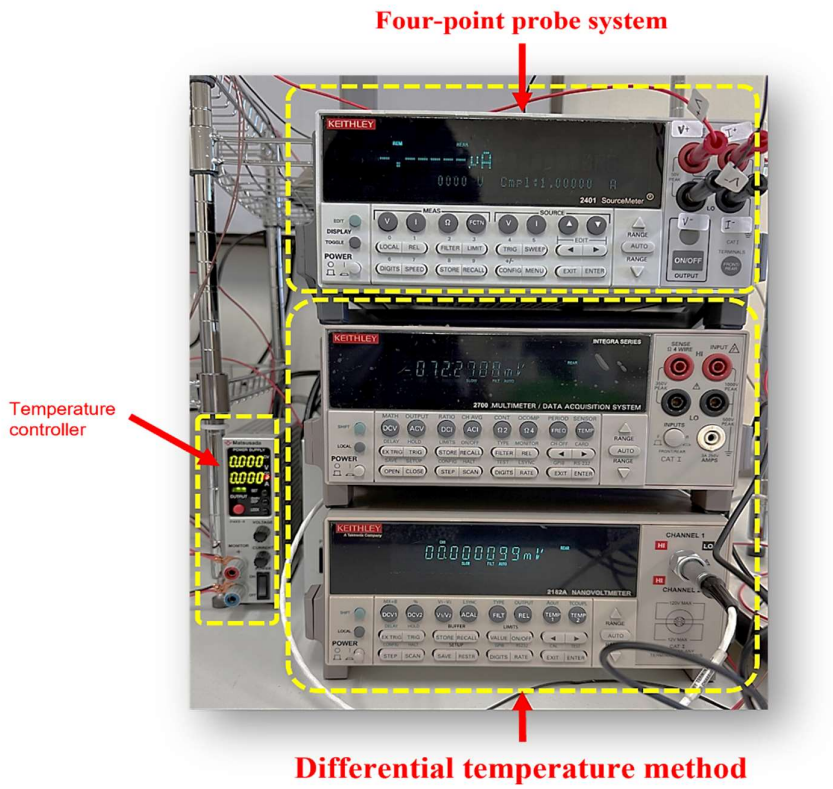


Figure- 17 Single integrated probe data acquisition system set-up

4.5 User Interface Design

To facilitate seamless communication with the integrated system, we utilize the Microsoft Excel 2007 version as the interface to retrieve data from the instruments via specific commands. This version of Excel serves as a platform to interact with the instruments, allowing us to send and receive commands to configure the measurement settings and acquire data from the Keithley 2182A, 2700, and 2401 instruments. The Microsoft Excel 2007 version acts as a convenient and user-friendly tool, enabling researchers to control the instruments, gather data, and perform data analysis all within a familiar spreadsheet environment. The integration of Excel with the instruments enhances the overall functionality and usability of the Data Acquisition System, using Visual Basic Applications (VBA) can be utilized to create a Data Acquisition System for measuring and analyzing data from the instruments. The first step is to set up the necessary variables to store the measurement data, such as the Seebeck coefficient and

electrical conductivity values. These variables will be used to store data collected from the Keithley 2182A, 2700, and 2401 instruments. Next, the VBA code will establish a connection with the instruments using the appropriate communication protocol (GPIB or USB). The code will send commands to the instruments via the GPIB-488 communication protocol to set up the desired measurement configurations, such as voltage ranges, current levels, and sample rates. The Keithley 2182A will be set up to read voltage measurements to calculate the Thermoelectromotive force(TEMF), while the Keithley 2700 will be configured to acquire voltage data for real-time monitoring. The Keithley 2401 will be employed to apply current to the fabric and measure the resulting voltage to determine the electrical conductivity.

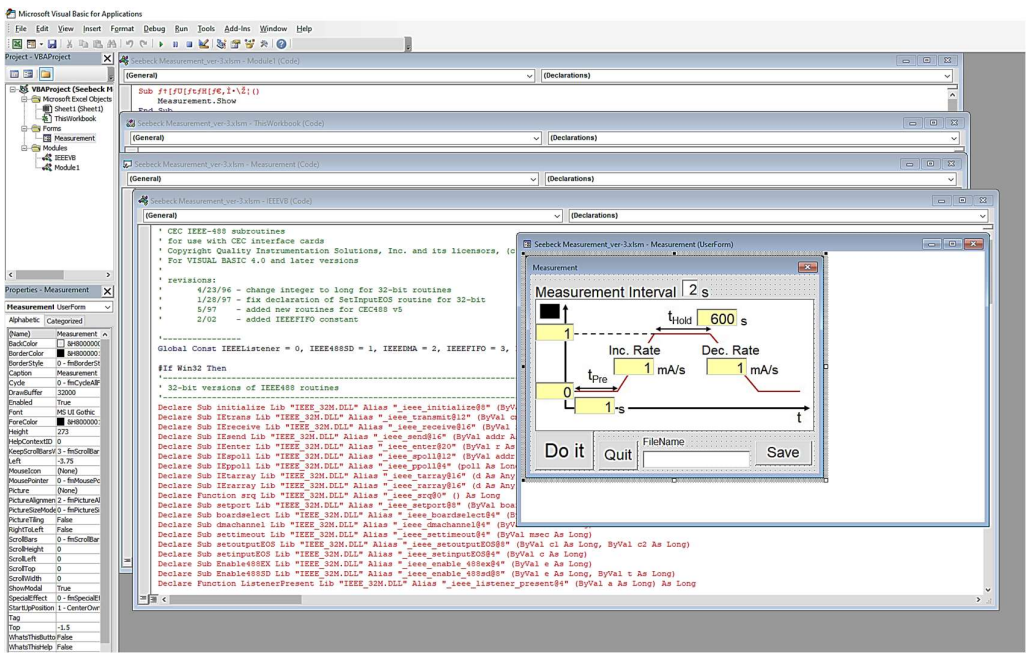


Figure- 18. Visual Basic Applications for Differential Temperature Method.

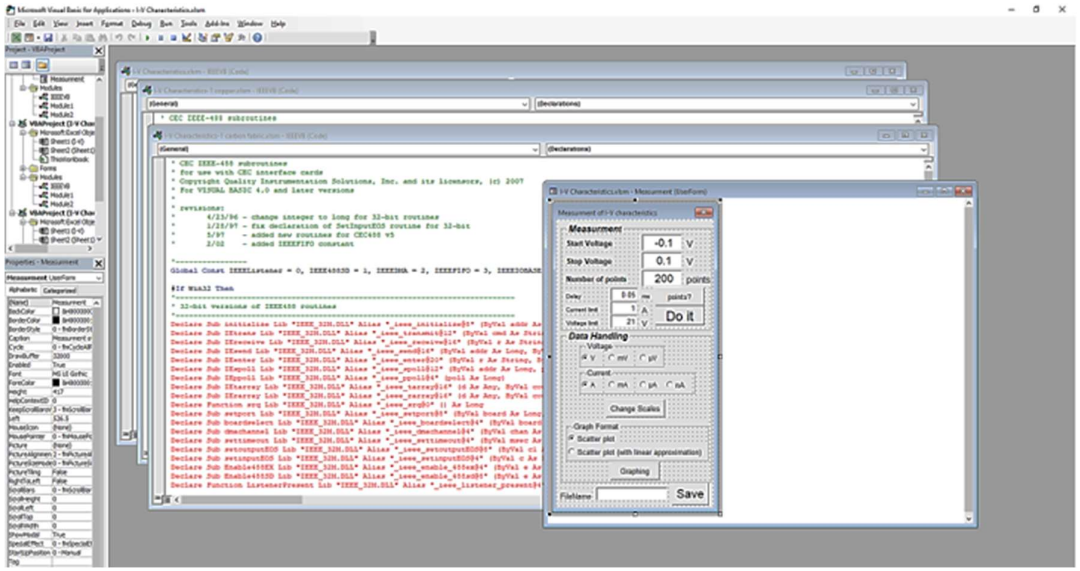


Figure- 19. Visual Basic Applications for Four-point Probe System.

The Data Acquisition System will incorporate loops to control the temperature using the Matsuda power supply[Fig-19]. The code will iteratively adjust the power supply settings to create the desired temperature gradients across the fabric sample. At each temperature point, the VBA code will trigger the Keithley 2182A to measure the voltage, which will be used to calculate the Seebeck coefficient. Simultaneously, the Keithley 2700 will record the voltage data for real-time monitoring, and An independent VBA code has been developed specifically for the electrical conductivity Data Acquisition using the Keithley 2401 instrument. This code enables the Keithley 2401 to carry out electrical measurements employing the four-point probe method. The four-point probe method is a precise technique for measuring electrical conductivity in materials, particularly flexible ones like conductive fabrics. The VBA code establishes a seamless connection between the Keithley 2401 and Microsoft Excel 2007, enabling efficient data retrieval and analysis. With this dedicated VBA code, researchers can easily configure the measurement parameters, initiate electrical conductivity measurements[Fig-20], and obtain accurate data with minimal effort. The integration of VBA code with the Keithley 2401 significantly enhances the overall

functionality of the Data Acquisition System, ensuring reliable and precise characterization of the electrical properties of flexible materials. Once all the data is collected, the VBA code will process and analyze the measurements to extract the Seebeck coefficient and electrical conductivity values. It will perform any necessary calculations or conversions to present the data in the desired format. The code will also create visualizations, such as graphs or charts, to aid in data interpretation and presentation.

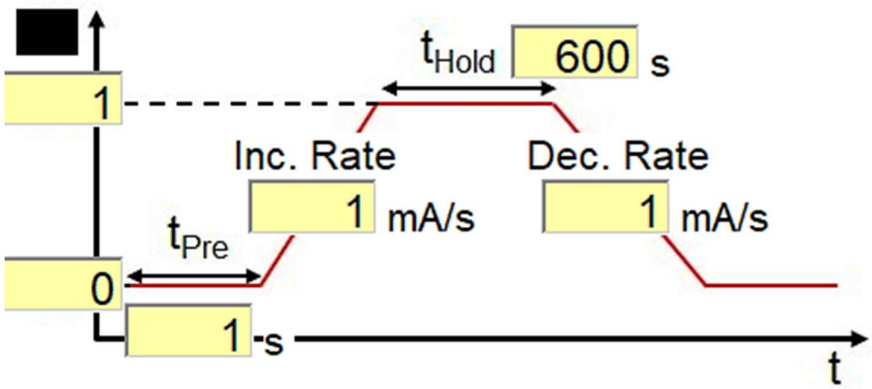


Figure- 19. Visual Basic Applications for temperature controlling system for the Differential temperature method.

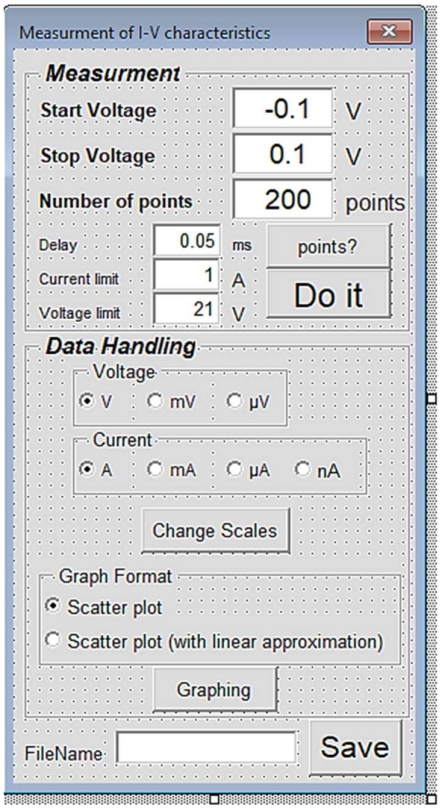


Figure- 20. Visual Basic Applications for Voltage Controlling System

for the Four-point probe system.

By employing VBA code in Excel, researchers and engineers can develop a custom Data Acquisition System that seamlessly integrates with the Matsuda power supply and Keithley instruments. This tailored approach allows for efficient and accurate measurements of the thermoelectric and electrical properties of flexible materials, providing valuable insights for various applications. The flexibility of Excel VBA enables users to adapt and extend the system's functionalities as needed, making it a powerful tool for advancing research and development in the field of flexible thermoelectric devices.

CHAPTER -5

Results and Discussion

5.1 Analysis of Flexible Materials

Analysis of flexible materials, such as conductive fabrics made with carbon fibers and nickel-copper alloys, presents exciting opportunities for various applications in wearable electronics and smart textiles. Carbon fabric, known for its exceptional mechanical properties and high electrical conductivity, is an ideal choice for flexible materials due to its lightweight nature and bendability. Nickel-copper fabrics, on the other hand, offer a unique combination of mechanical flexibility and excellent electrical conductivity, making them suitable for a wide range of electronic and sensing applications. In this analysis, we explore the electrical, and thermal, properties of both carbon fabric and nickel-copper fabric-based conductive materials to understand their potential for use in energy harvesting and wearable electronics.

Additionally, investigating the thermal properties of carbon fabric and nickel-copper fabric conductive materials is crucial. The Seebeck coefficient measurement is a key parameter for understanding their thermoelectric properties. By inducing controlled temperature gradients across the materials and measuring the resulting voltage differences, the Seebeck coefficient can be determined. The Seebeck coefficient signifies the materials' ability to convert thermal energy into electrical energy, making it a critical factor in thermoelectric energy harvesting. Conductive fabrics with high Seebeck coefficients have the potential to harvest energy from the human body or the surrounding environment, offering sustainable power sources for wearable electronics and IoT devices.

Carbon fabric (CAF)

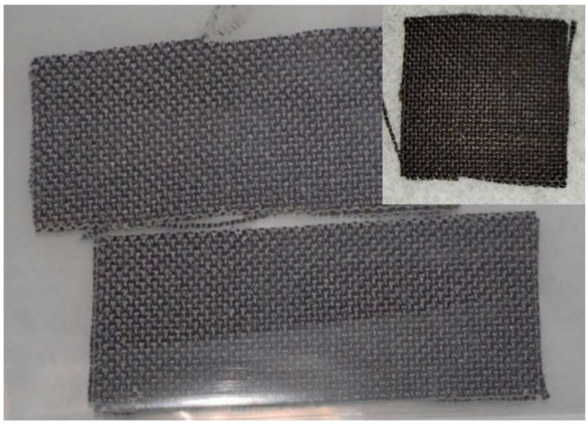


Figure-21.5x2 and 2x2Bare CAF.

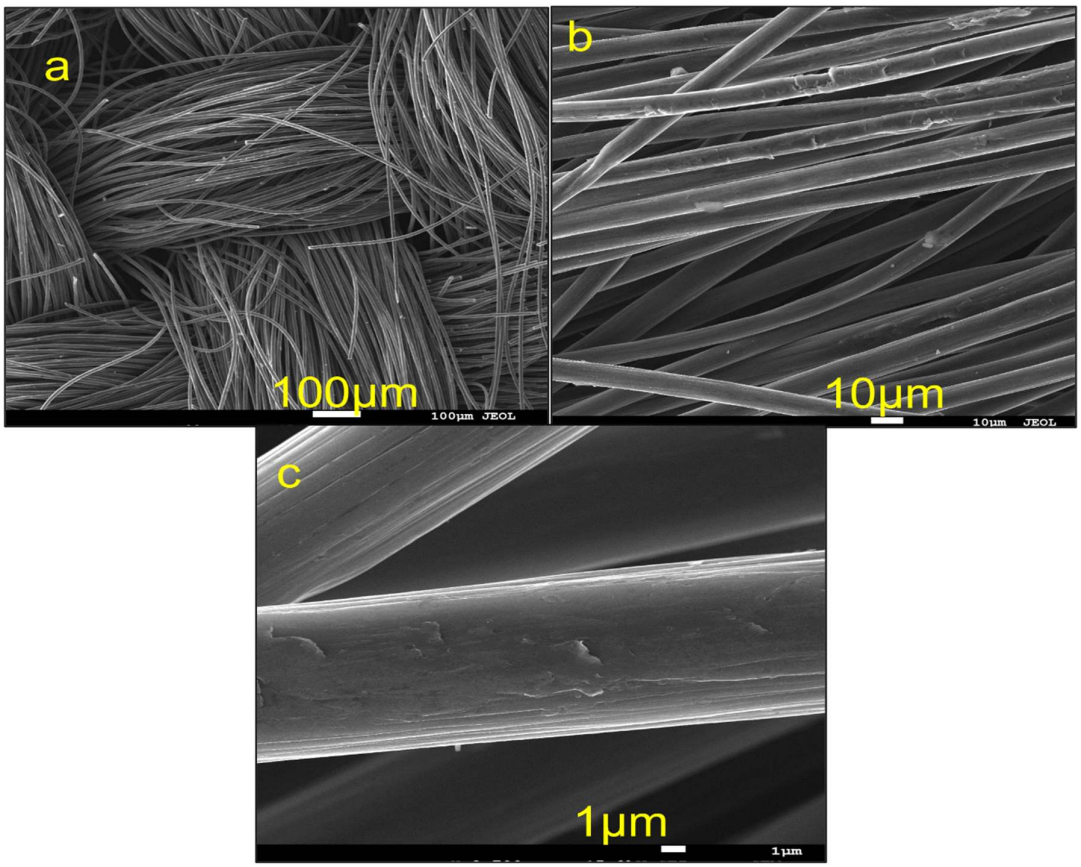


Figure-22. SEM images (100µm ,10µm ,1µm)of the Bare CAF.

Nickel-copper fabric (NCF)

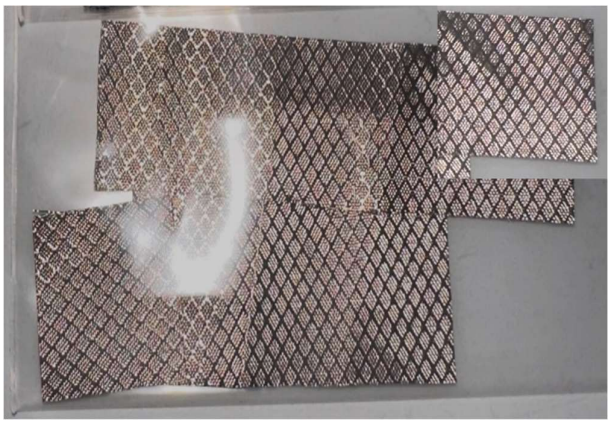


Figure-23 5x2and 2x2 Dimond shape NCF.

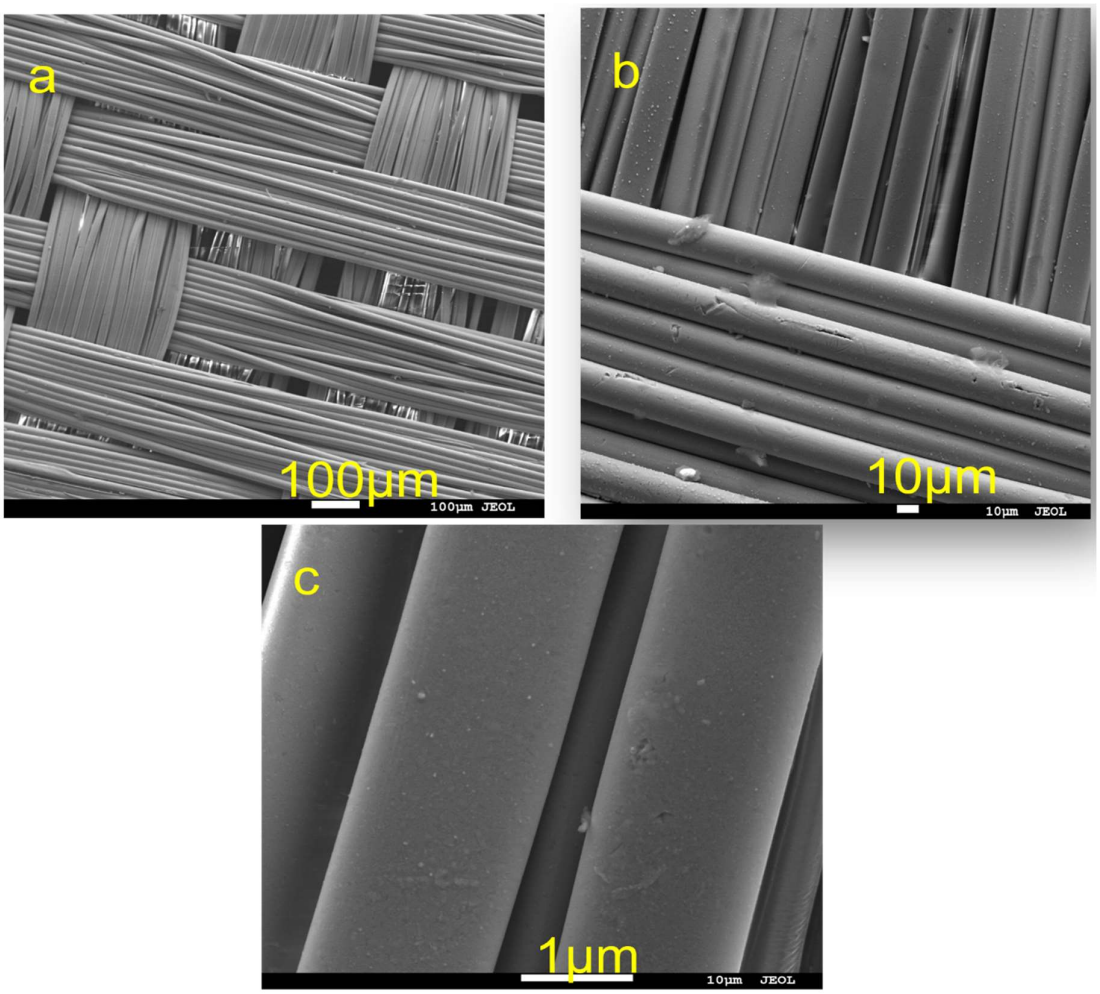


Figure 24. SEM images(100µm ,10µm ,1µm) of the bare diamond shape NCF.

5.2 Seebeck Coefficient Measurements

Seebeck coefficient measurements are a crucial aspect of characterizing the thermoelectric properties of materials. The Seebeck coefficient represents a material's ability to generate a voltage in response to a temperature gradient, a phenomenon known as the Seebeck effect. The measurements involve creating controlled temperature gradients across the material[Fig-25] and observing the voltage differences that arise. To perform Seebeck coefficient measurements, a differential temperature setup is typically employed. This involves placing the material between two temperature-controlled probes, where one end is heated and the other is cooled. The resulting voltage difference is measured using electrodes, and the Seebeck coefficient is calculated as the ratio of the generated voltage to the temperature difference. We used a temperature control system that allowed us to create a temperature difference of 30°C. We conducted the measurement of the fabric under this 30°C temperature difference condition with the correction of the 0.2μV/k. The results of the measurement are provided in Table 2.

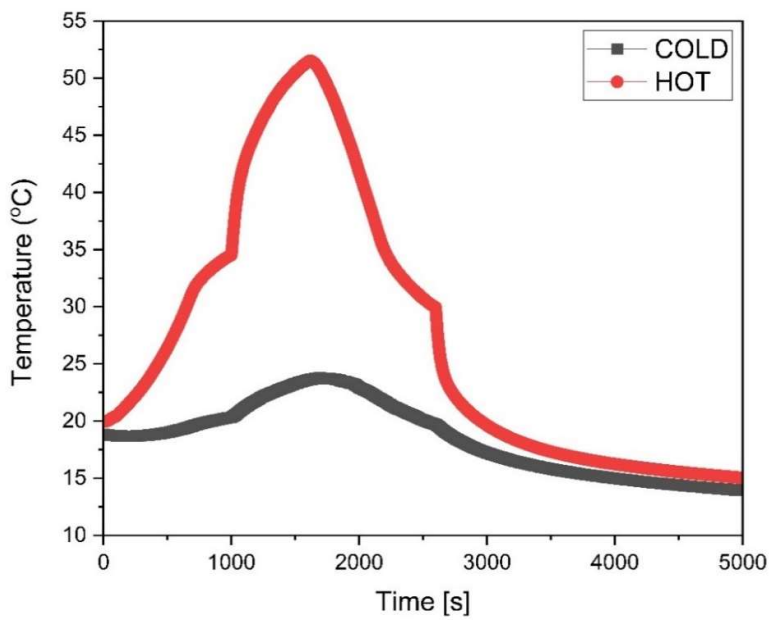


Figure -25 shows the relation between the Temperature difference and Time.

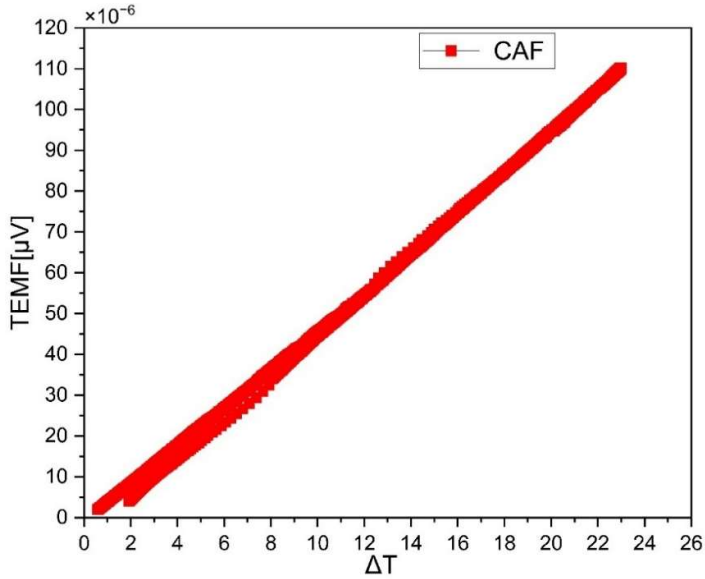


Figure -26 shows the Thermoelectric motive force (TEMF) depends on the Temperature difference of the CAF.

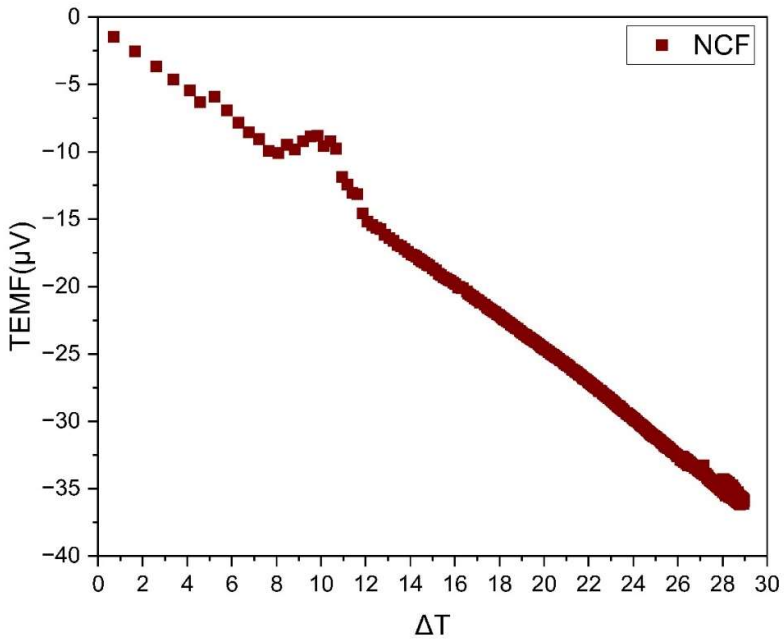


Figure -27 shows the Thermoelectric motive force(TEMF) depends on the Temperature difference of the NCF.

Original fabric	Parameters	NCF	CAF
Reported work	Seebeck coefficient (μV/K)	-2.7	5.05
Current work	Seebeck coefficient (μV/K)	-2.5(±0.2)	4.9(±0.2)

Table -2. Seebeck coefficient comparison of the previous work and current work.

5.2 Electrical Conductivity Measurements

An aspect of analyzing flexible conductive fabrics is characterizing their electrical conductivity. Electrical measurements, such as the four-point probe method, can accurately determine the electrical conductivity of carbon fabric materials. This technique involves passing a known electrical current through the fabric and measuring the voltage drop across the two sets of closely spaced probes. By knowing the dimensions of the fabric and the measured voltage and current, electrical conductivity can be calculated. electrical conductivity measurements are essential for understanding the material's electrical performance and assessing its suitability for specific electronic applications.

We employed the Vander Pauw method, utilizing a commercial Hall effect measurement system, to determine the electrical conductivity of carbon fabrics (CAF). During the measurements, we encountered an issue with the sharp probe of the Hall instrument, as it tended to penetrate through the fabric due to its wavy nature. To address this, we made modifications by adopting the apex probe system [see Fig-28], which helped alleviate the probe penetration problem. Despite resolving the probe penetration issue, we faced difficulties in reading the data accurately due to machine vibrations. To overcome this challenge, we designed and implemented a closed cap made of non-conductive material, which effectively secured the fabric and apex probe in place, ensuring proper and consistent contact between the fabric and the probe. Through the Hall measurement with the modified setup, we successfully obtained the

electrical conductivity of the carbon fabric, which was determined to be 2.86e17/cm^2. It is worth noting that the Hall measurement system which we are using does not have an integrated temperature modification capability. However, given the scope to examining the electrical conductivity across different temperature ranges, we opted for the four-probe measurement technique with temperature variation control.

By utilizing the four-probe measurement method, we could investigate the electrical conductivity of the carbon fabric at various temperatures, providing valuable insights into its temperature-dependent behavior. This allowed us to obtain a comprehensive understanding of the material's electrical properties under diverse thermal conditions.

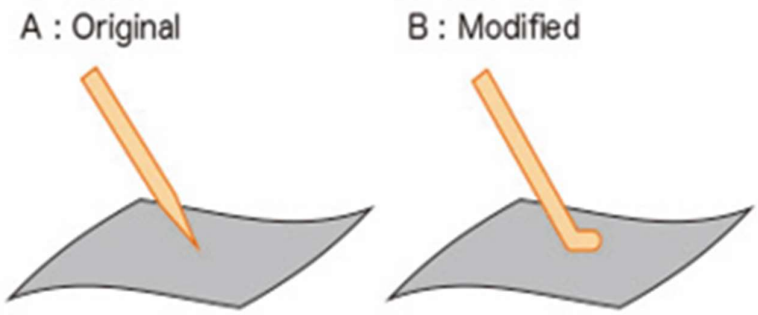
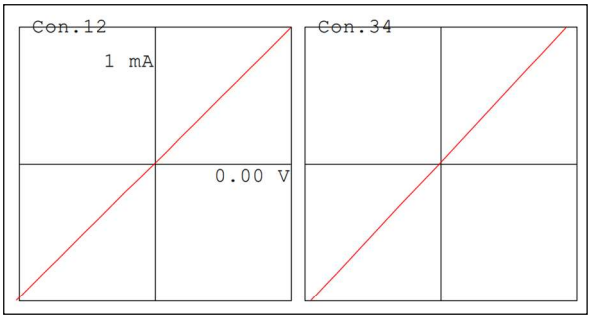
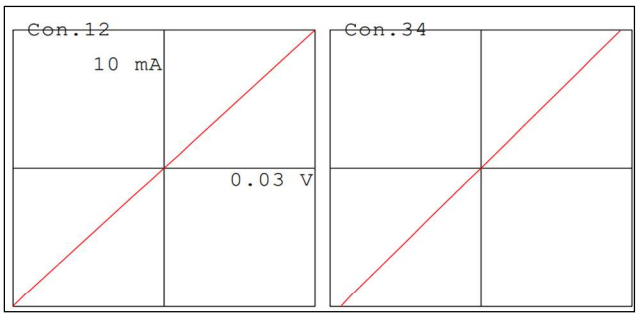


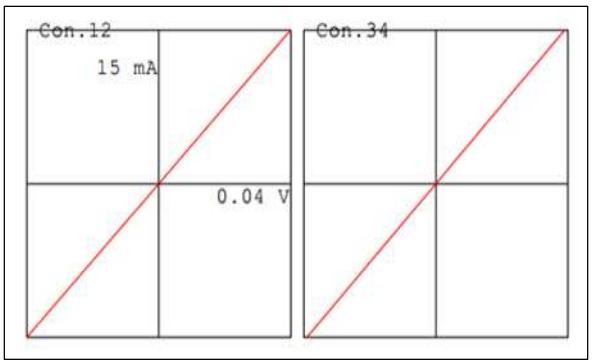
Figure -28.Modification of probes from (A)sharp apex to (B)obtuse apex.



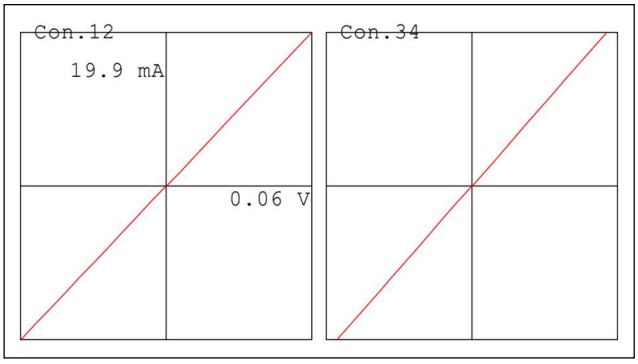
1mA



10mA

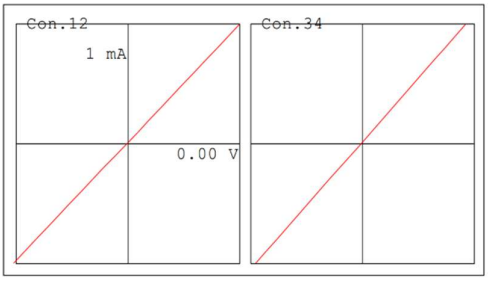


15mA

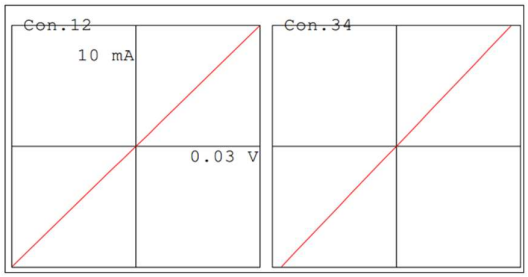


19.9mA

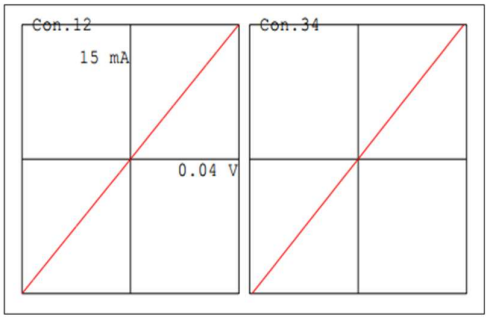
I-V characteristics of a carbon fabric measured with obtuse apex probes.



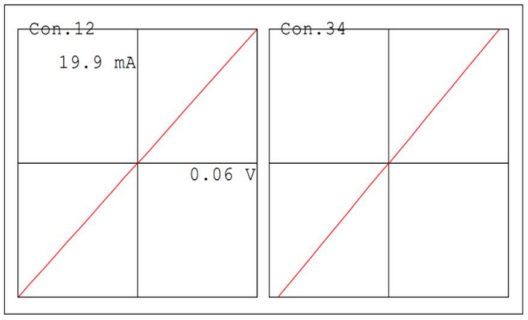
1mA



10mA



15mA



19.9mA

Figure 29. I-V characteristics of a carbon fabric measured with obtuse apex. Probes.

In the four-probe method, we proceeded to measure the electrical properties of the fabric. To ensure the accuracy of our measurements, we first conducted a calibration test using copper as a reference material (see Fig-30). During this calibration, we carefully examined the I-V (current-voltage) curve of copper and determined the slope and interception values to establish a reliable baseline for our measurements.

Following the successful calibration, we proceeded to measure the I-V curve of the carbon fabric (CAF) with a correction factor of ±0.1 nA. This correction allowed us to account for any potential deviations or uncertainties in our measurements and ensure the reliability of the data obtained. The measured I-V curve of the CAF is depicted in (Fig-31), providing valuable insights into its electrical characteristics and conductivity. By employing the four-probe method and conducting the necessary calibration, we obtained accurate and trustworthy data for further analysis and understanding of the fabric's electrical behavior.

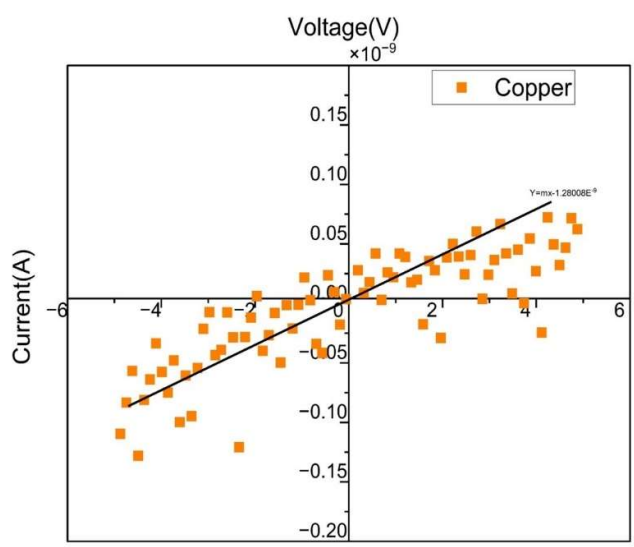


Figure-30. I-V curve of the 2mm thick Copperplate.

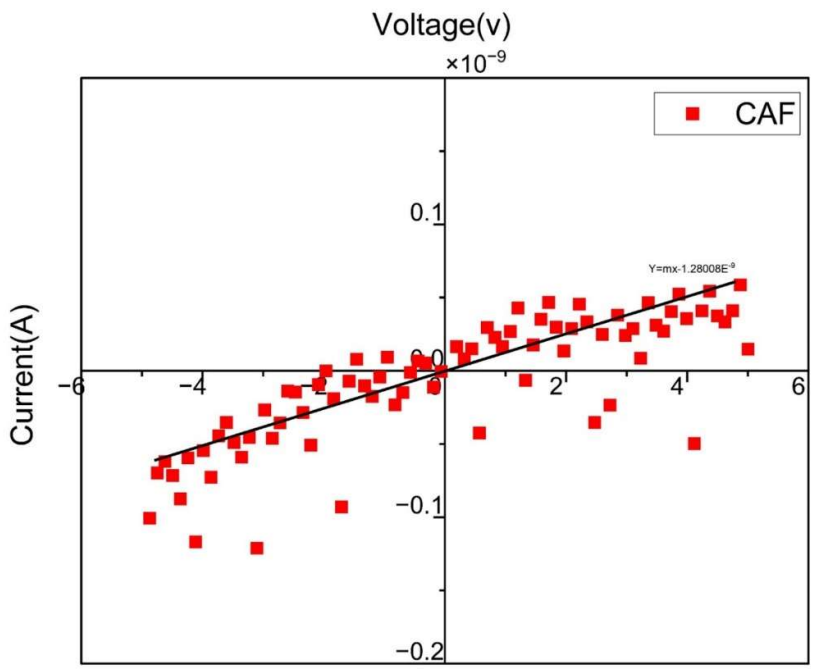


Figure-31. I-V curve of the 0.25mm of CAF.

In our experiment, we achieved the design and development of the Seebeck coefficient and electrical conductive of the integrated probe for flexible devices.

CHAPTER -6

Applications and Future Enhancements

6.1 Potential Applications

The Design and development of the integrated instrument for measuring the Seebeck coefficient and electrical conductivity of flexible materials has supports numerous potential applications. Here are some possible applications which can characterized by using home made integrated instrument set up

Thermoelectric Power Generation: The instrument enables researchers to identify and optimize flexible materials with high Seebeck coefficients, which are crucial for efficient thermoelectric power generation. These materials can be utilized in portable and wearable devices to convert waste heat into electricity, powering various electronic components.

Personalized Temperature Control: The instrument's ability to measure Seebeck coefficients allows for the optimization of flexible materials for personalized climate control applications. By converting temperature differences between the body and the environment into electrical energy, these materials can provide localized heating or cooling for enhanced comfort.

Energy Harvesting from Human Body: Flexible thermoelectric materials integrated into wearable garments can harvest body heat to generate electricity. The instrument helps assess the thermoelectric performance of these materials, enabling the development of self-powered wearable devices, such as smartwatches or health monitoring devices.

Smart Textiles: Smart textiles with integrated sensors and electronic components rely on flexible materials with specific electrical conductivity properties. The instrument assists in selecting and optimizing conductive fabrics for applications in health monitoring, motion tracking, and interactive textiles.

Environmental Monitoring: Flexible sensors made from conductive fabrics can be used for environmental monitoring applications. The instrument aids in evaluating the

electrical conductivity of these sensors, which are employed for detecting various environmental parameters like humidity, temperature, and pollution levels.

Flexible Electronics: The electrical conductivity measurement capability of the instrument is valuable for developing flexible electronics. It aids in characterizing conductive fabrics used in flexible circuits, touchscreens, and other electronic components integrated into bendable and wearable devices.

Structural Health Monitoring: The instrument can contribute to the development of flexible sensors used for structural health monitoring in buildings, bridges, and infrastructure. The Seebeck coefficient measurement helps assess the thermoelectric performance of these sensors, ensuring accurate and reliable data collection.

Internet of Things (IoT) Devices: With the growing demand for IoT devices, the instrument plays a crucial role in selecting suitable flexible materials for the fabrication of energy-efficient and reliable IoT sensors and actuators.

Flexible Energy Storage: The instrument's capability to measure electrical conductivity in flexible materials is useful for developing flexible energy storage devices. These materials can be employed in flexible supercapacitors or batteries for various portable and wearable applications.

Automotive Applications: The instrument's ability to evaluate the thermoelectric and electrical properties of flexible materials is relevant for automotive applications. It can aid in developing thermoelectric generators that convert waste heat from vehicle exhaust into electricity, contributing to improved fuel efficiency and reduced emissions.

From the above possible application, we are focused to work on Energy Harvesting from Human Body by using conductive fabrics which act as Flexible thermoelectric materials integrated into wearable garments that can harvest body heat to generate electricity. The integrated instrument we constructed will help assess the

thermoelectric performance of these materials, enabling the development of self-powered wearable devices, such as smartwatches or health monitoring devices.

6.2 Possible Improvements and Expansions

In the future, Possible Improvements and Expansions by integration of MATLAB and 3D printing technology can significantly enhance the Design and development of the Seebeck coefficient and electrical conductivity integrated instrument for flexible materials. Firstly by 3D Printed Instrument Parts, Utilizing 3D printing technology for fabricating instrument parts offers several advantages. Custom-designed components can be quickly prototyped and manufactured, allowing for faster iterations and improvements in the instrument's overall design. 3D printing also enables the creation of complex geometries and structures, ensuring precise alignment of probes and enhanced accuracy in measurements. Moreover, 3D printing allows for the production of lightweight and portable instrument parts, making the system more convenient for field applications.

Secondly, By incorporating MATLAB into the instrument's design, researchers can leverage its powerful data analysis and visualization capabilities. MATLAB can be used to process the data collected from Keithley 2182A, 2700, and 2401 instruments, allowing for more advanced analysis of the Seebeck coefficient and electrical conductivity measurements. Additionally, MATLAB programming capabilities enable the development of custom algorithms and automation scripts to streamline data processing and increase the instrument's efficiency. MATLAB's ability to communicate with external devices, such as the Keithley instruments, facilitates seamless data acquisition and control. By integrating the Seebeck coefficient and electrical conductivity systems into one controlling software within MATLAB, researchers can efficiently manage both measurements simultaneously. The software can offer a user-friendly interface where users can select the desired measurement mode, set parameters, and view real-time data visualization for both thermoelectric and electrical measurements. This integrated software streamlines the characterization

process and provides a comprehensive analysis of the flexible materials' properties. Researchers can develop MATLAB scripts to send commands to the instruments, set specific measurement parameters, and retrieve data in real time. This integration ensures efficient data collection, minimizing manual intervention and potential errors.

Multi-Modal Characterization: The combination of Seebeck coefficient and electrical conductivity measurements in one system offers a unique advantage. Researchers can perform multi-modal characterization of flexible materials, gaining deeper insights into their thermoelectric and electrical properties simultaneously. This comprehensive analysis helps in identifying materials with desirable properties for specific applications, leading to more optimized and efficient designs. Automated Measurement Sequences: MATLAB's programming capabilities enable the creation of automated measurement sequences. Researchers can define protocols that involve performing a series of Seebeck and electrical conductivity measurements under different conditions. This automation saves time, improves reproducibility, and allows for systematic exploration of various material samples.

In conclusion, integrating MATLAB and 3D printing technology into the instrument's design enhances its capabilities in terms of data analysis, customization, and automation. The combined controlling software and multi-modal characterization offer researchers a comprehensive and efficient tool for studying the thermoelectric and electrical properties of flexible materials. These improvements not only streamline the characterization process but also provide valuable insights for developing advanced flexible materials with applications in energy harvesting, wearable electronics, and more.

6.3 Conclusion

The development of flexible and wearable thermoelectric devices holds immense potential for various applications, such as energy harvesting, personalized climate control, and wearable electronics. These devices offer the advantage of seamlessly

integrating into wearable garments and accessories, enabling comfortable and unobtrusive usage. The use of materials like polymers, composites, conductive fabrics, and thin films ensures excellent flexibility and durability, making them suitable for daily wear. Despite the progress made in this field, challenges remain in improving device efficiency, flexibility, and heat management. Collaborative efforts among materials scientists, engineers, and designers are essential to overcome these hurdles and fully unleash the potential of flexible and wearable thermoelectric devices. By harnessing the abundant thermal energy from the human body and optimizing the properties of flexible materials through an integrated instrument capable of measuring the Seebeck coefficient and electrical conductivity, this research contributes to advancing the field and opens up exciting opportunities for future innovations in wearable technology and energy harvesting.

Flexible and wearable thermoelectric devices have a lot of exciting possibilities! They can be used in energy harvesting, climate control, and wearable electronics. These devices can easily fit into our clothes and accessories, making them comfortable to wear every day. However, there are some challenges that we need to overcome. We need to improve the efficiency, flexibility, and heat management of these devices. To do that, we need to work together with materials scientists, engineers, and designers. By collaborating, we can make these devices even better and unlock their full potential for various applications in wearable technology and energy harvesting. Our project focused on developing an integrated instrument that can measure the Seebeck coefficient and electrical conductivity of conductive fabrics. This instrument is really useful for researchers and engineers who want to analyze and improve conductive fabrics used in wearable electronics and other applications. With this instrument, we can monitor and understand how these fabrics behave in real-time, and that helps us optimize their performance.

We measured the Seebeck coefficient to understand how well these fabrics can convert thermal energy into electrical energy. This is important for energy harvesting.

Fabrics with high Seebeck coefficients can generate electricity from body heat or the environment, which is great for self-powered wearable devices like smartwatches and health monitors. Another important part of our project was measuring the electrical conductivity of carbon fabric. This helps us understand how well the fabric can conduct electricity. By using a modified setup, we successfully measured the electrical conductivity under different temperature conditions. This data is essential for designing electronic components that can be integrated into flexible materials.

our project has achieved its main goal of developing an integrated instrument for flexible materials. This instrument is a valuable tool for researchers and engineers working on wearable electronics and energy harvesting. By working together and continuously improving these devices, we can create more efficient and sustainable technologies that will benefit us all.

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List of publications and conferences

1. **S.T. Venkata Sai Varma**, Pravarthana Prabhu, Hiroya Ikeda, Pandiyarasan Veluswamy. "Thermoelectric generator on cellulose fabric coated Bi_2Te_3 for the wearable application". presented poster presentation at the "International Conference On Advanced Materials And Mechanical Characterization (ICAMMC-2021)" (VIRTUAL MODE, 2- 4 December 2021), organized by the Department of Physics and Nanotechnology and Department of Mechanical Engineering, SRM Institute Of Science And Technology in association with the IISc, IIT Delhi, IIT Madras, IIT Hyderabad, IIT Indore, Indian Institute Of Metals Chennai Chapter, ASM International Chennai Chapter, Indian Ceramic Society, Indian Physics Association, and American Ceramic Society India Chapter.
2. **S.T. Venkata Sai Varma**, Pravarthana Prabhu, Pandiyarasan Veluswamy. "Cellulose-based Bi_2Te_3 coated fabric for enhanced thermoelectric performance ". presented poster presentation at the "International Conference On Advanced Materials And Mechanical Characterization (ICAMMC-2021)" (VIRTUAL MODE, 2- 4 December 2021), organized by the Department of Physics and Nanotechnology and Department of Mechanical Engineering, SRM Institute Of Science And Technology Technology in association with the IISc, IIT Delhi, IIT Madras, IIT Hyderabad, IIT Indore, Indian Institute Of Metals Chennai Chapter, ASM International Chennai Chapter, Indian Ceramic Society, Indian Physics Association, and American Ceramic Society India Chapter.
3. **S.T. Venkata Sai Varma**, Pandiyarasan Veluswamy, Hiroya Ikeda, "Thin film development of Bi_2Te_3 on Cellulose fabric for gas sensor application" at Takayanagi Symposium, Shizuoka University, Japan.
4. **S.T. Venkata Sai Varma**, Erdenetuul Dorjsuren , R. pandeeswari ,V. Balamuragan , Pandiyarasan Veluswamy , Ikeda Hiroya. "Thin film development of Bi_2 Te_3 on Cellulose Fabric for hybrid application" at ISFAR-SU2023 symposium, Shizuoka University, Japan.